

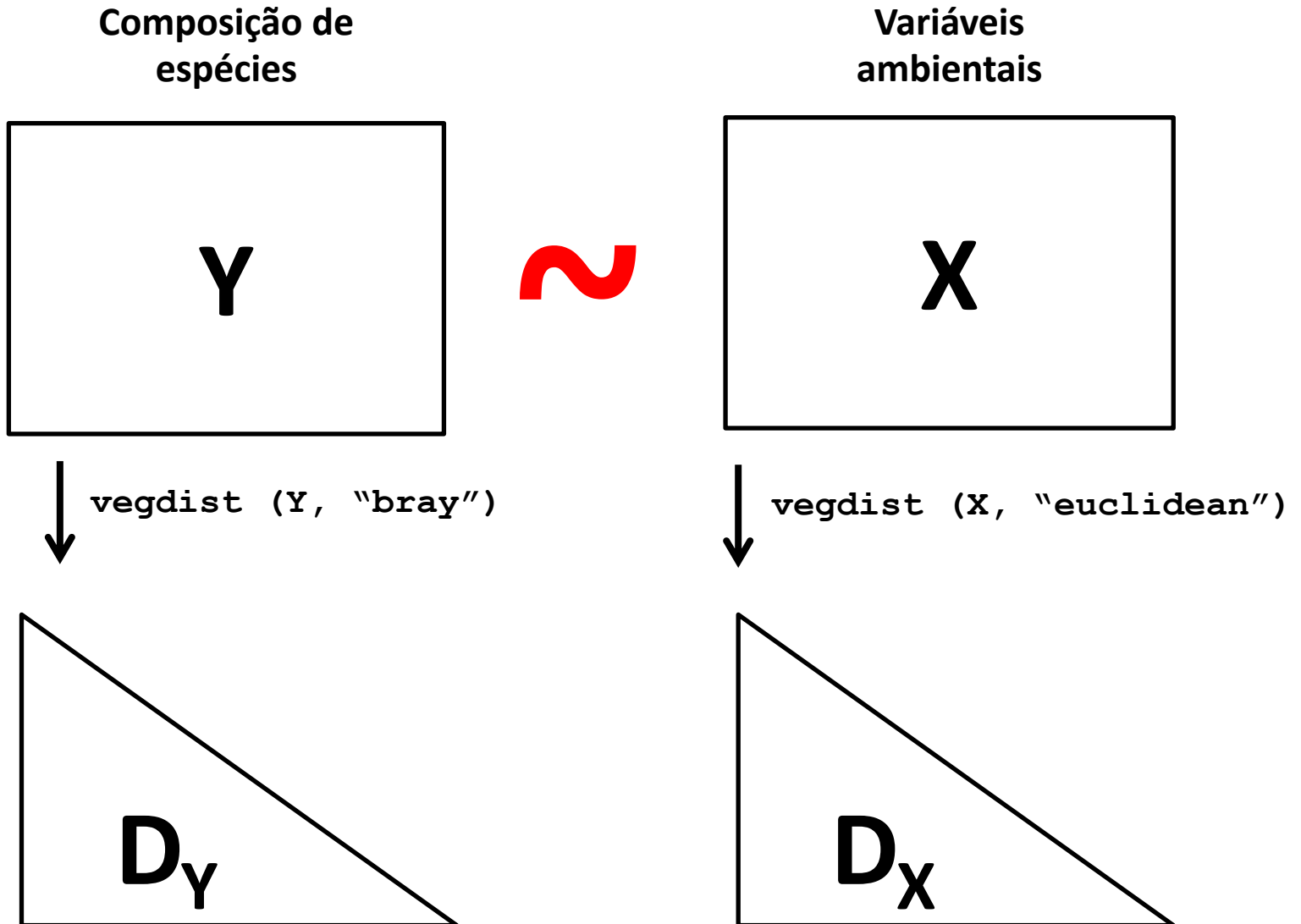


# Aula 6: Métodos para testar relações entre variáveis e diferença entre grupos

p. 597-612 do Cap. 10 Legendre &  
Legendre

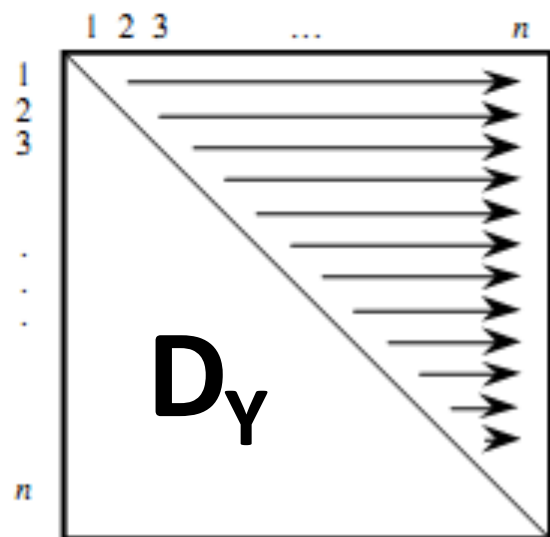
- Mantel
- Mantel parcial
- RMR
- Procrustes/Protest
- PERMANOVA
- betadisper+permdisp

# Teste de Mantel

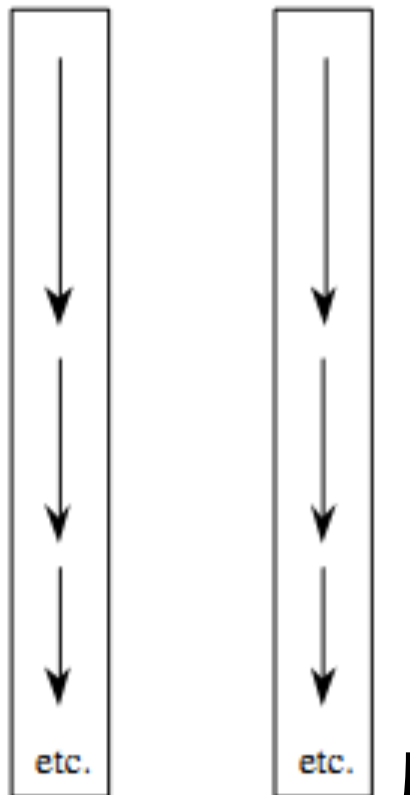




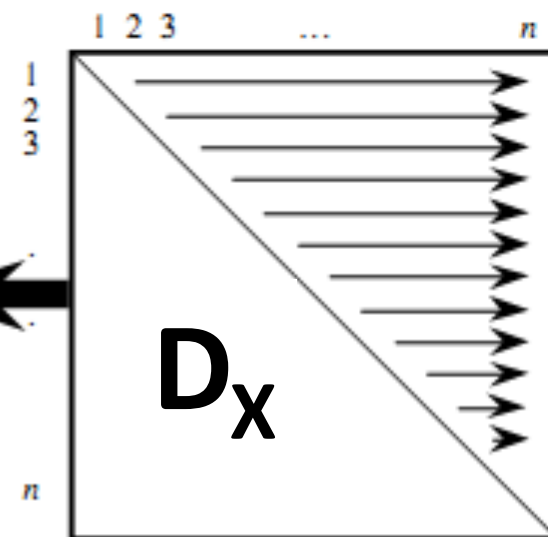
$S$  or  $D$  computed from  
a first data table



Unfold the  $S$  or  $D$  matrices



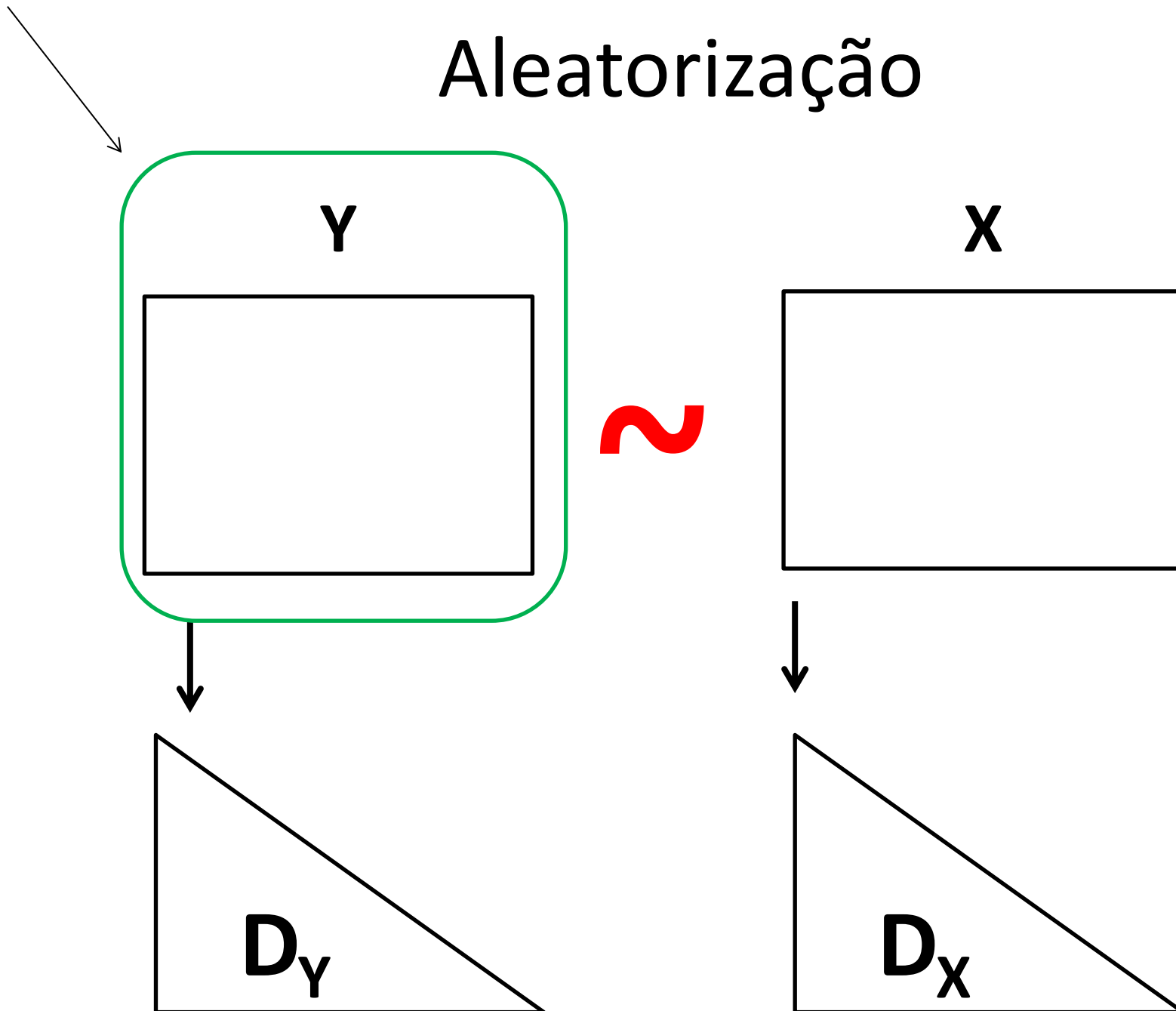
$S$  or  $D$  computed from  
a second data table



Produto cruzado  
dos vetores

Estatística de  
Mantel:  $r_M$

# Aleatorização



# Mantel parcial

- Relaciona dois conjuntos de matrizes de distância, controlando para uma terceira
- Lógica similar à da db-RDA, só que usando matrizes de distância
- A ordem das matrizes importa
- Útil para testar relação entre distância genética e distância ambiental, controlando para distância geográfica

# Vantagens

Possibilidade de usar qualquer coeficiente de distâncias

Qualquer tipo de dado pode ser usado para construir a matriz de distância, inclusive mistura de dados

Permite usar mais de uma matriz (testes parciais)

## Desvantagens

Ter de transformar os dados brutos (matriz quadrada) em distância, com perda inerente de informação

Interpretação é dada no espaço de distâncias

Perda da informação das observações originais

Baixo poder (Erro Tipo II) e alto Erro Tipo I

### 3. Two types of $R^2$ statistics and SS

The  $R_M^2$  statistic of the Mantel test differs from the  $R^2$  of correlation, regression, or canonical analysis. Demonstration:

$R^2$  of multiple regression and RDA :  $R^2 = \frac{SS(\hat{\mathbf{Y}})}{SS(\mathbf{Y})}$

Its denominator is  $SS(\mathbf{Y}) = \sum_{i=1}^n \sum_{j=1}^p (y_{ij} - \bar{y}_j)^2 = \frac{1}{n} \sum_{h=1}^{n-1} \sum_{i=h+1}^n D_{hi}^2$  \*

The  $R_M^2$  of the Mantel test is the square of the Mantel correlation ( $r_M$ ).

Its denominator is  $SS(\mathbf{D})$ , not  $SS(\mathbf{Y})$ :

$$SS(\mathbf{D}) = \sum_{h=1}^{n-1} \sum_{i=h+1}^n (D_{hi} - \bar{D})^2 = \sum_{h=1}^{n-1} \sum_{i=h+1}^n D_{hi}^2 - \frac{(\sum \sum D_{hi})^2}{n(n-1)/2}$$

number of distances

These two statistics are irreducible to one another.

## 4. Two types of SS – An ecological example

Consider a matrix of species presence-absence at four sites. There is variation among the sites (beta diversity).

Data **Y**

|        | Sp.1 | Sp.2 | Sp.3 | Sp.4 | Sp.5 |
|--------|------|------|------|------|------|
| Site 1 | 1    | 1    | 0    | 0    | 0    |
| Site 2 | 1    | 0    | 1    | 0    | 0    |
| Site 3 | 1    | 0    | 0    | 1    | 0    |
| Site 4 | 1    | 0    | 0    | 0    | 1    |

There's variation in sp composition between sites

**SS(Y) of the data = 3.000**

SS(from the Jaccard *D*) =

$$6*(0.667^2)/4 = 0.667$$

**D** = [1 – Jaccard similarity]

|        | Site 1 | Site 2 | Site 3 | Site 4 |
|--------|--------|--------|--------|--------|
| Site 1 | 0      | 0.667  | 0.667  | 0.667  |
| Site 2 | 0.667  | 0      | 0.667  | 0.667  |
| Site 3 | 0.667  | 0.667  | 0      | 0.667  |
| Site 4 | 0.667  | 0.667  | 0.667  | 0      |

The numbers are equal because they only differ in one species

**SS(D) in the upper triangle = 0.000**

Mantel would find there's no variation in the data

The variation of the distances in matrix **D** does not provide any information about the variation in the **Y** data.



## *5. Basic assumptions of the Mantel test: Do they hold in spatial analysis?*

1. Mantel correlation: hypothesis of linearity (or at least of monotonic relationship) of the point dispersion in the **D–D** graph.
2. When  $D_1$  is small,  $D_2$  is small; when  $D_1$  is large,  $D_2$  is large. The distribution of values in the **D–D** graph is homoscedastic.

Are these assumptions verified in data on which ecologists and geneticists carry out spatial analysis?

⇒ Let us examine spatially autocorrelated (SA) simulated data.

⇒ Advantage of simulated data: we know for sure the range of the spatial autocorrelation.

Mantel serve para análises espaciais?

# COMPARISON OF PERMUTATION METHODS FOR THE PARTIAL CORRELATION AND PARTIAL MANTEL TESTS

PIERRE LEGENDRE\*

## CONCEPTS & SYNTHESIS

EMPHASIZING NEW IDEAS TO STIMULATE RESEARCH IN ECOLOGY

*Ecological Monographs*, 75(4), 2005, pp. 435–450  
© 2005 by the Ecological Society of America

## ANALYZING BETA DIVERSITY: PARTITIONING THE SPATIAL VARIATION OF COMMUNITY COMPOSITION DATA

PIERRE LEGENDRE,<sup>1</sup> DANIEL BORCARD, AND PEDRO R. PERES-NETO<sup>2</sup>

### MOLECULAR ECOLOGY RESOURCES

*Molecular Ecology Resources* (2010) 10, 831–844

doi: 10.1111/j.1755-0998.2010.02866.x

### METHODOLOGICAL ADVANCES - INFERENCE OF SPATIAL STRUCTURE

## Comparison of the Mantel test and alternative approaches for detecting complex multivariate relationships in the spatial analysis of genetic data

PIERRE LEGENDRE\* and MARIE-JOSÉE FORTIN†

## Methods in Ecology and Evolution



British Ecological Society

*Methods in Ecology and Evolution* 2015, 6, 1239–1247

doi: 10.1111/2041-210X.12425

## Should the Mantel test be used in spatial analysis?

Pierre Legendre<sup>1\*</sup>, Marie-Josée Fortin<sup>2</sup> and Daniel Borcard<sup>1</sup>

- Like the previous results, these new simulations show that the power of the Mantel test is always lower than that of canonical analysis (RDA).
- Our results also show that spatial variation is, at best, weakly captured by direct regression of a response distance (or dissimilarity) matrix on a geographic distance matrix.
- None of the transformations of the distances that we tried increased the performance of the Mantel test.
- On the RDA side, representation of the spatial relationships among sites by Moran's eigenvector maps (MEM) produces a much more powerful test than the use of simple geographic coordinates or in the form of a polynomial.

## 9. Conclusion

Researchers like the Mantel test because it is simple to use. One simply has to type: `mantel(D1, D2)` to obtain a p-value.

1. The Mantel test is not appropriate to test for the presence of spatial structures in raw survey data, for several reasons.

1.1. The null hypothesis ( $H_0$ ) of the test of correlation between vectors of raw data is different from  $H_0$  of a test involving dissimilarities.

1.2. The statistics used in these two tests are different and cannot be reduced to one another.

1.3. The Mantel correlation assumes that, in a **D–D** plot, the relationship among points is linear and homoscedastic. That is not the case for spatially-structured data, except when the range of the spatial correlation extends to the whole surface under study or beyond.

## 9. Conclusion

2. If one still applies the Mantel test to spatially-structured data, its power is always lower than that of RDA using Moran's eigenvector maps (MEM) to represent spatial relationships among sites.

Our simulations showed that the Mantel  $R^2$  is always much lower than the  $R^2$  produced by MEM analysis. It should not be interpreted as the proportion of the variance of  $\mathbf{Y}$  explained by  $\mathbf{X}$ .

=> Analysis by spatial eigenvectors (MEM, AEM) produces output richer than a p-value, which is the only output of a Mantel test. In particular, the ordination triplots display the relationships among sites, species, and MEM eigenvectors. They also allow one to produce maps of the fitted values at different spatial scales.

3. The Mantel test is inappropriate to test the correlation between raw data vectors or matrices, irrespective of the fact that these vectors are spatially-structured or not, because the Mantel statistic is inappropriate to answer that type of question; the test would have low power. See simulations in Legendre et Fortin (2010).

## 9. Conclusion

4. Our main conclusion is that the Mantel test should only be used to answer questions that, in the application field, clearly and solely concern *relationships between distances*. Such questions are rarely found in ecology and genetics.

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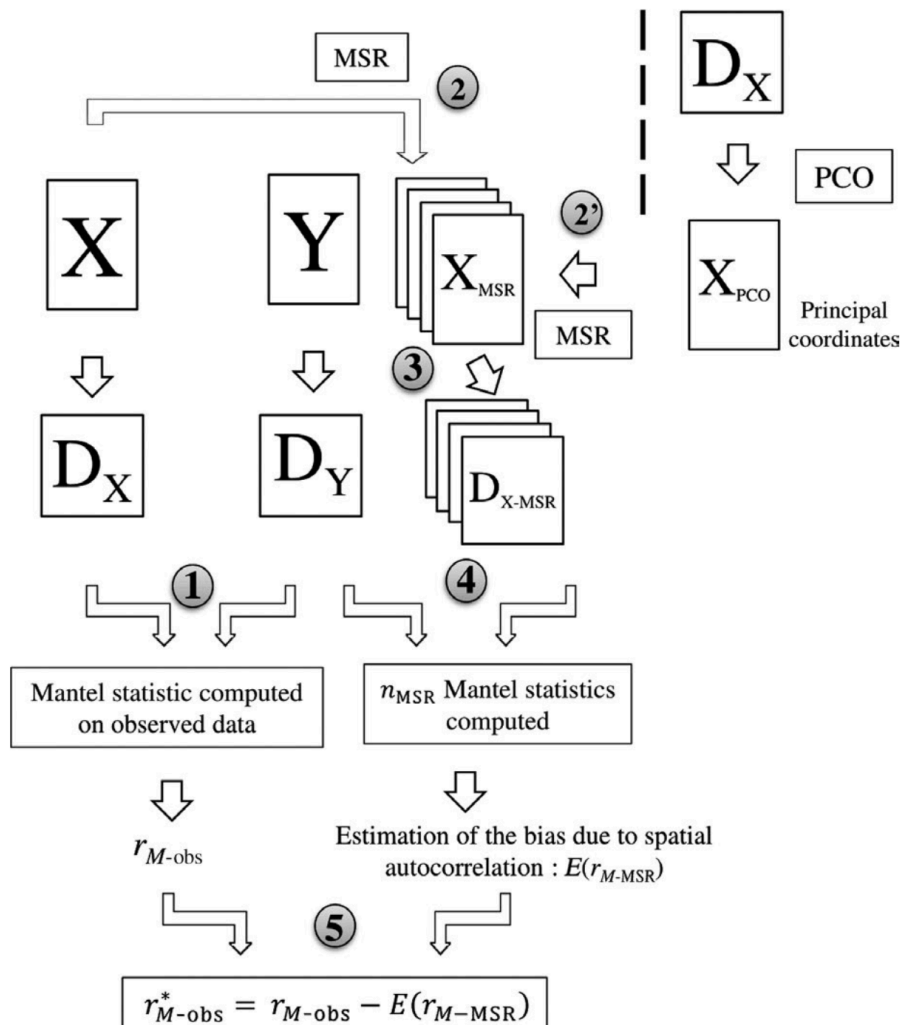
It should not be used to answer questions derived from questions that originally concerned the raw data, from which dissimilarities have been computed.

5. Who wants to use a test that has low power (low capacity to detect an effect), instead of a test that has high power?

# Testing the Mantel statistic with a spatially-constrained permutation procedure

Julie Crabot<sup>1\*</sup>  | Sylvie Clappe<sup>2\*</sup> | Stéphane Dray<sup>2</sup>  | Thibault Datry<sup>1</sup> 

*Methods Ecol Evol.* 2019;10:532–540.



Usa um novo  
procedimento para  
gerar modelos nulos  
espacialmente restritos

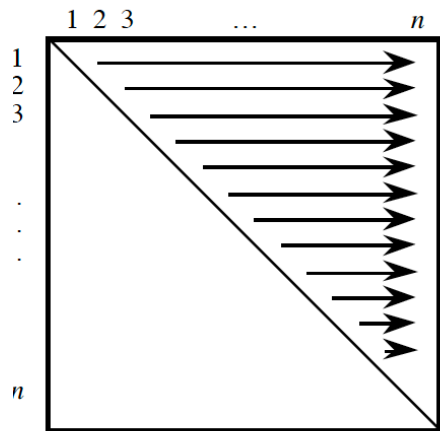


# Regressão Múltipla de Matrizes de distância

# O que é e como funciona?

- Muito similar ao Mantel, mas ao invés de calcular correlação de Pearson, calcula regressão
- Também permite várias matrizes preditoras
- Portanto requer variável preditora e resposta
- Fornece  $R^2$

**Dependent matrix  $D_Y$**

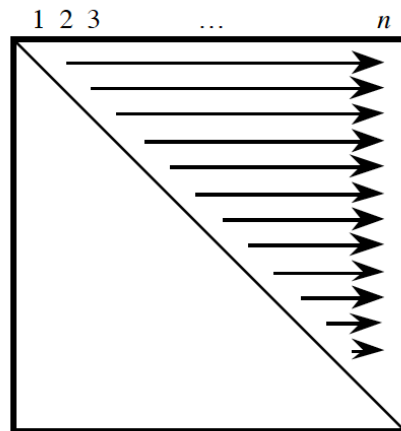


Unfold  
this matrix

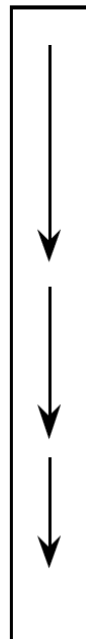


Vector  $y$

**Independent matrix  $D_{X1}$**

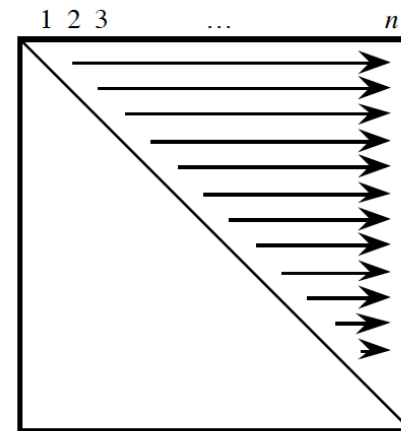


Unfold  
this matrix



Vector  $x_1$

**Independent matrix  $D_{X2}$**



Unfold  
this matrix



Vector  $x_2$

**Etc.**

**Etc.**

# Generalized Dissimilarity Modelling (GDM)

*Syst. Biol.* 51(2):331–363, 2002

# Mapping Spatial Pattern in Biodiversity for Regional Conservation Planning: Where to from Here?

SIMON FERRIER

*Diversity and Distributions, (Diversity Distrib.)* (2007) 13, 252–264

**BIODIVERSITY  
RESEARCH**



**Using generalized dissimilarity modelling to analyse and predict patterns of beta diversity in regional biodiversity assessment**

Simon Ferrier<sup>1\*</sup>, Glenn Manion<sup>1</sup>, Jane Elith<sup>2</sup> and Karen Richardson<sup>3</sup>

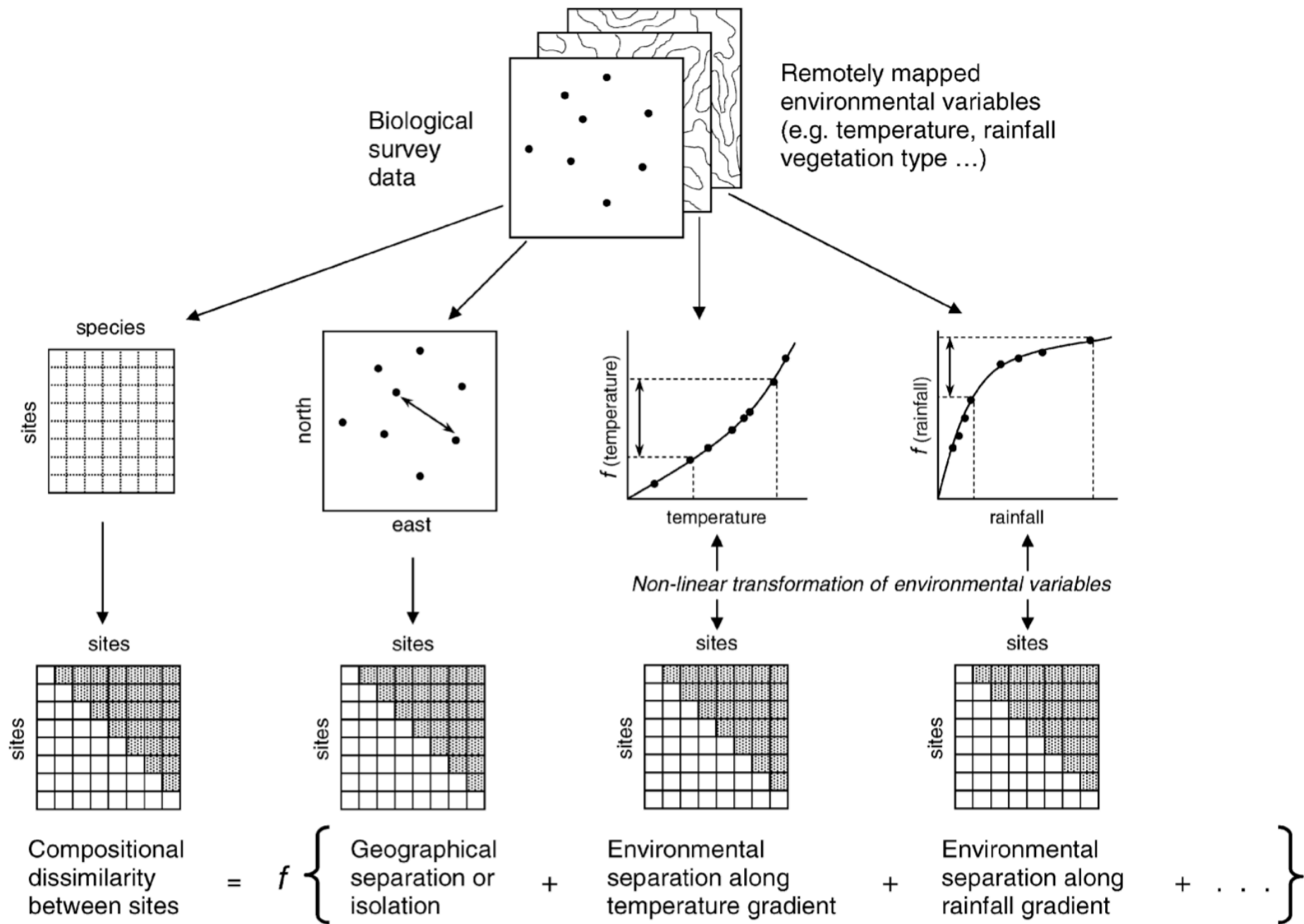
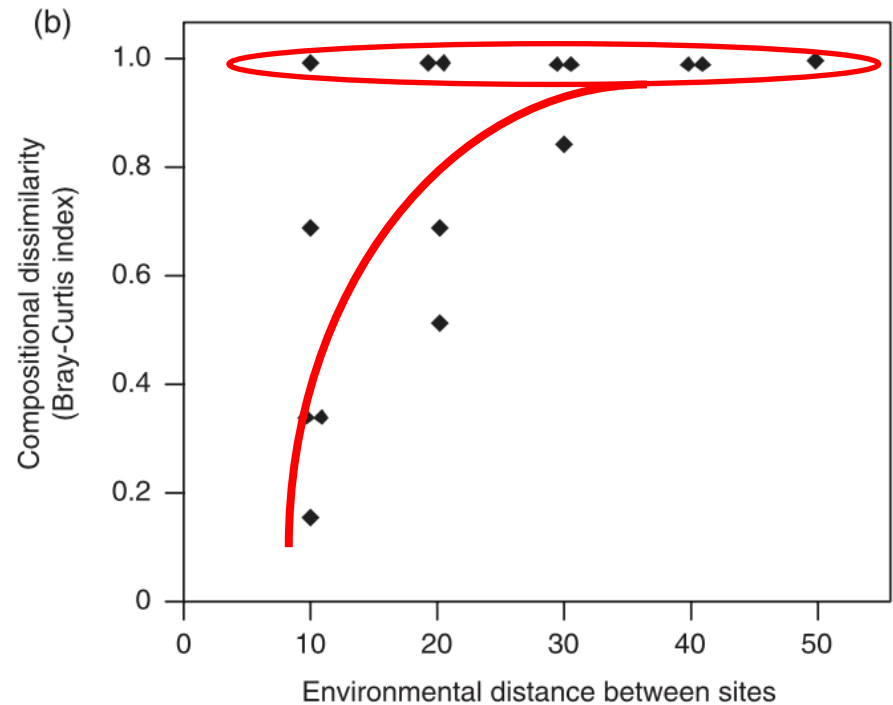
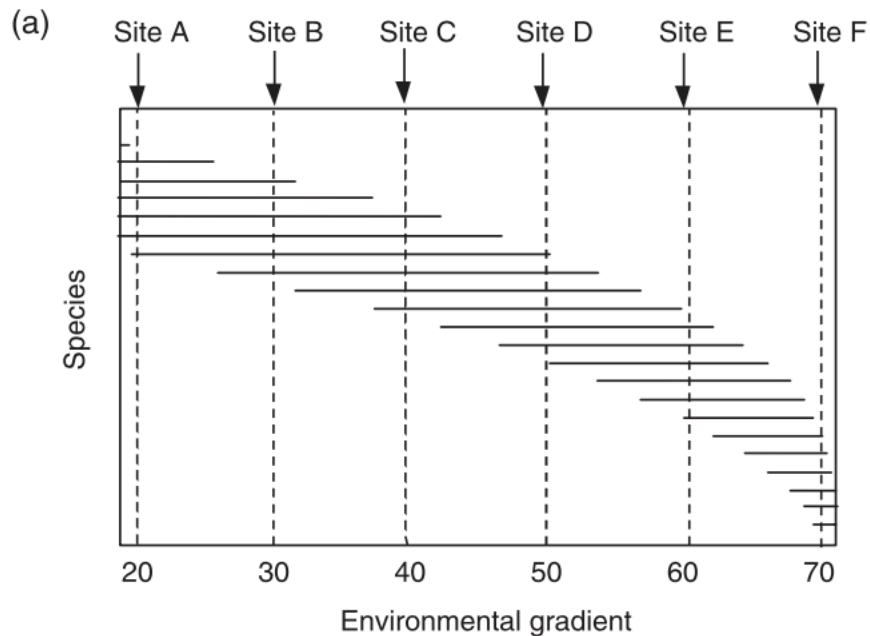


Figure 8. A diagrammatic illustration of the GDM approach to modelling compositional dissimilarity

# Generalized Dissimilarity Modelling

- Extensão da MRM
- Acomoda relações não-lineares
- Bom desempenho em modelar matrizes de distância com saturação
- Aplicações em modelagem de comunidades, mapeamento



**Figure 1** Hypothetical example illustrating the problems discussed in the text relating to the application of linear matrix regression to large-scaled ecological data sets. (a) Species' distributions (horizontal lines) in relation to a hypothetical environmental gradient, with six survey sites positioned along this gradient. (b) Plot of compositional dissimilarity vs. environmental distance for all possible pairwise combinations of the six survey sites.



# Análise PROcrustes

In Greek mythology, **Procrustes** was a son of **Poseidon** and a **rogue**. He invited travellers to spend the night with him, then tied them down to an iron bed and either cut off their limbs if they were taller than the bed, or stretched the victims if they were too short, till they fitted in.

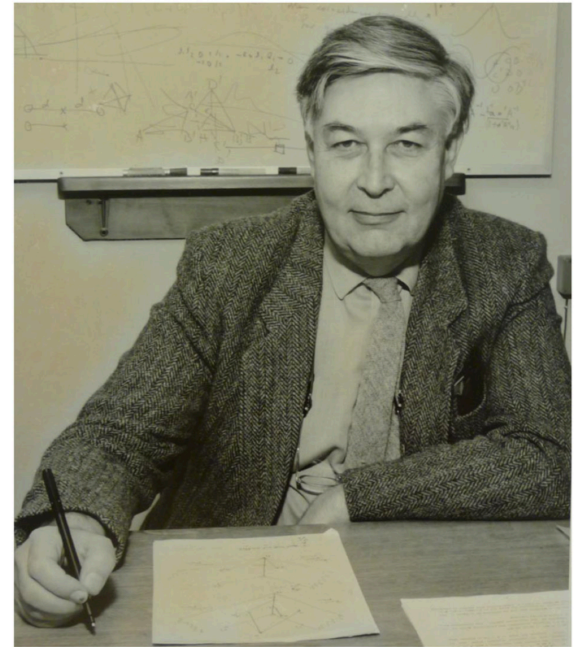


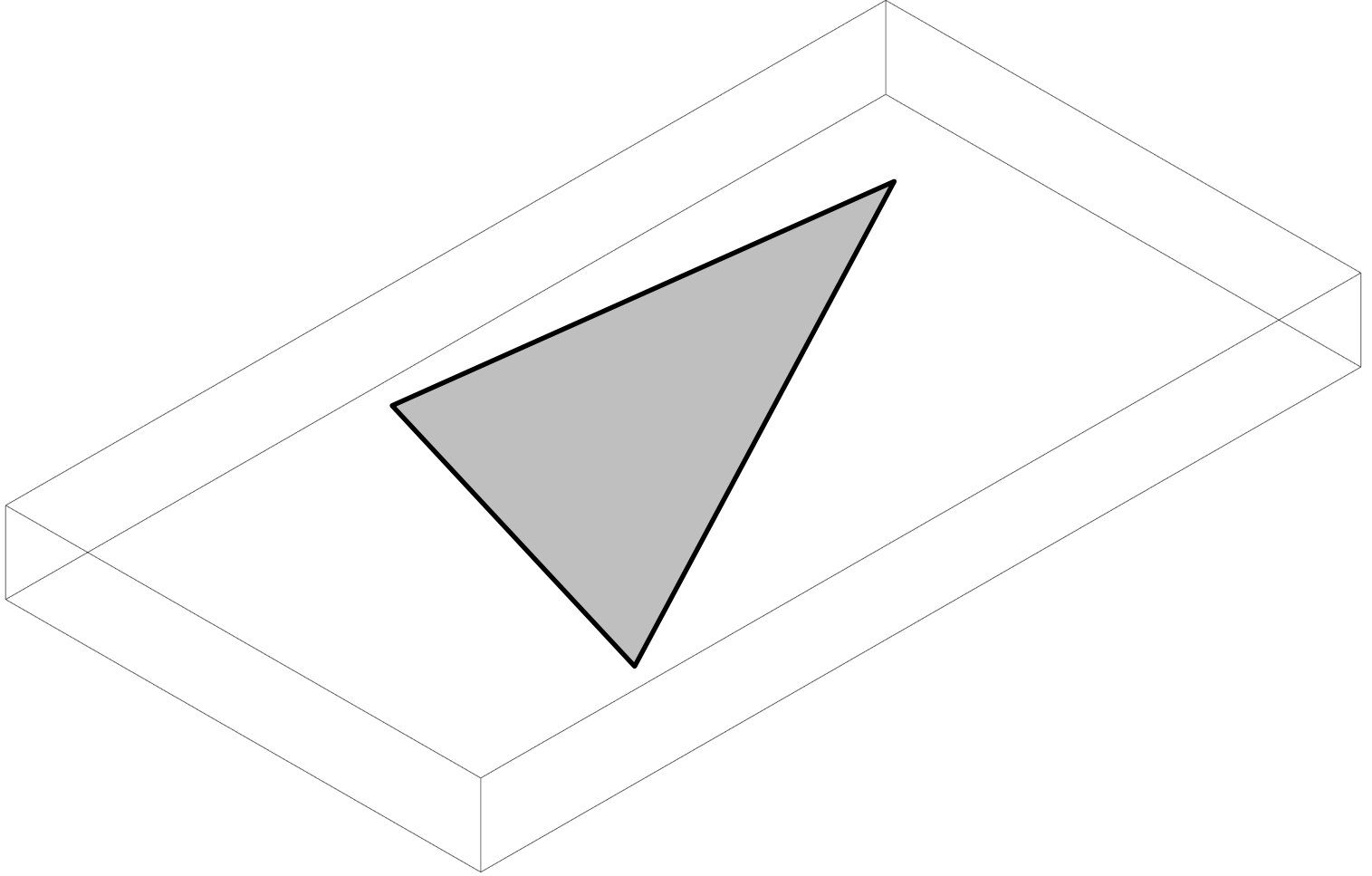
Figure 1. John Gower, circa 1985.

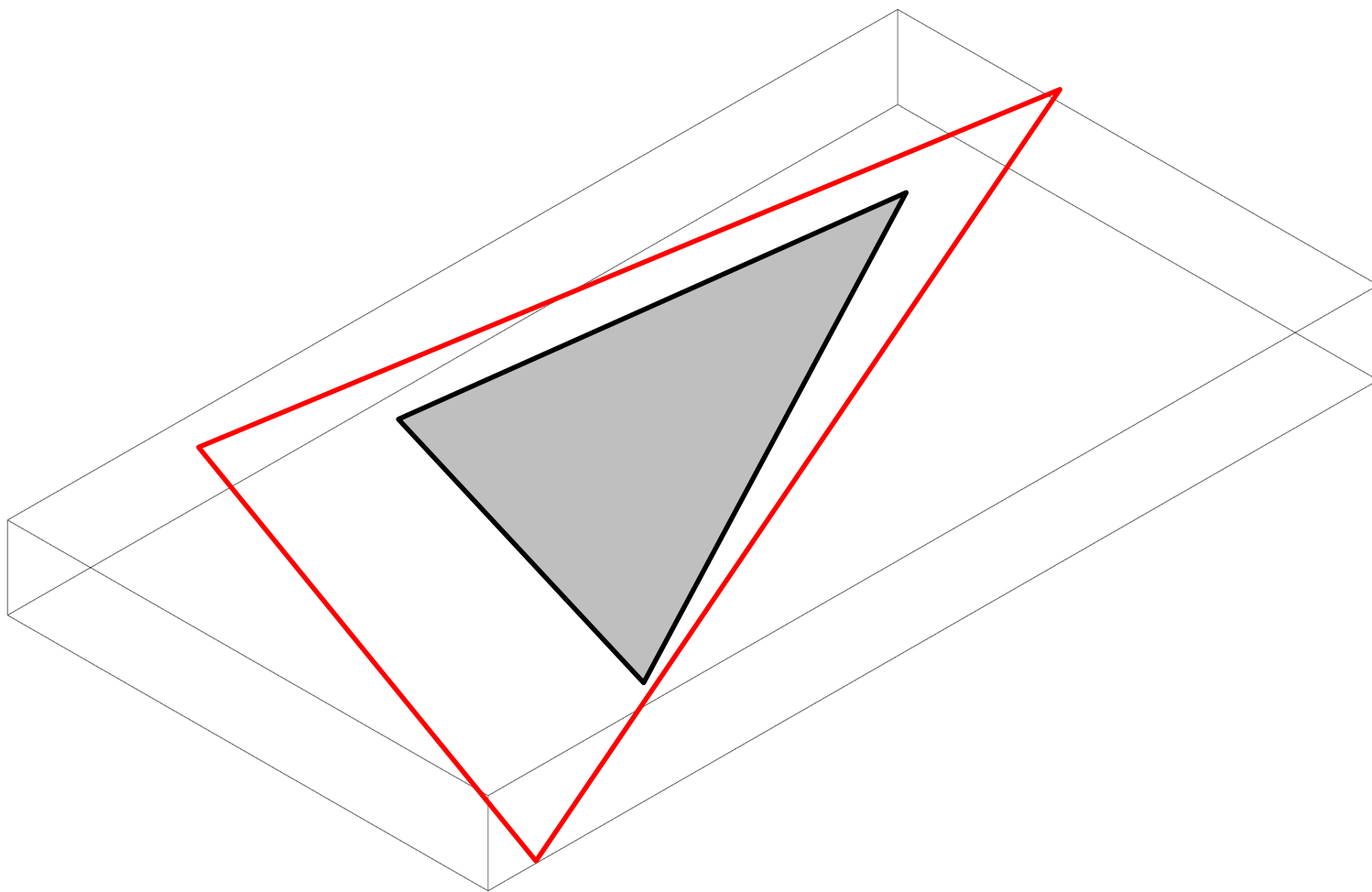
Tradução de procrusto (em português) = “O esticador”

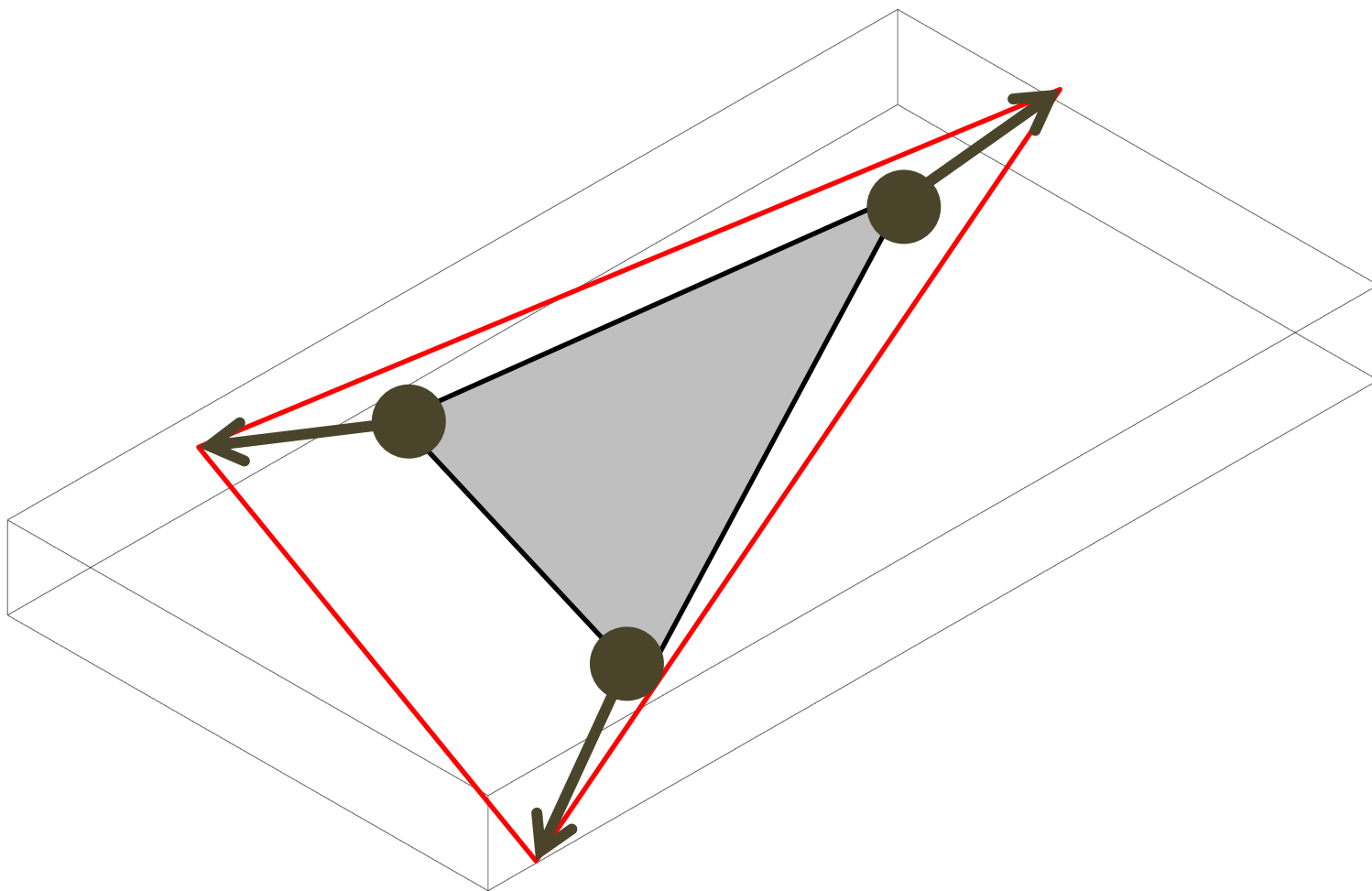
# Procrustes e Teseu

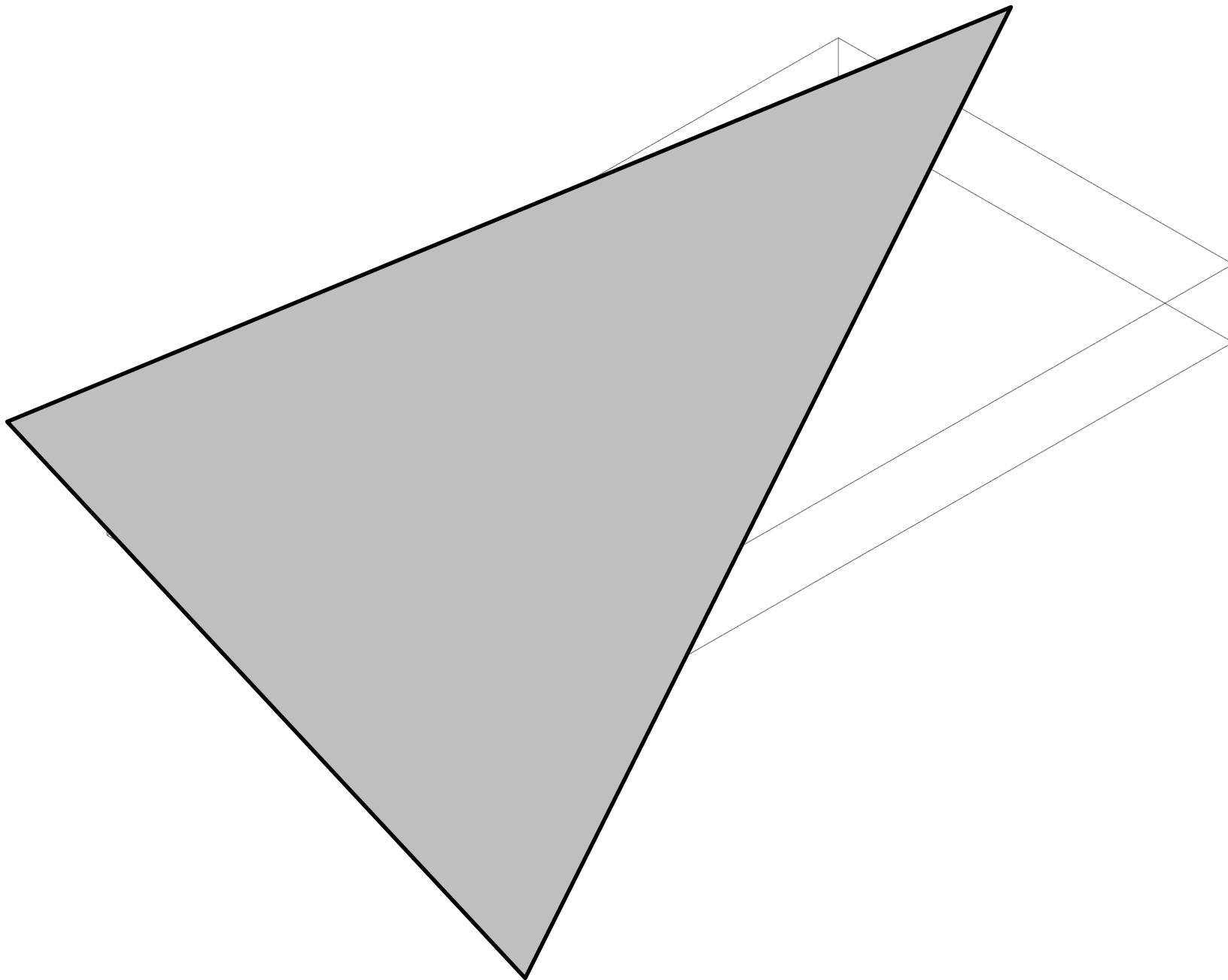
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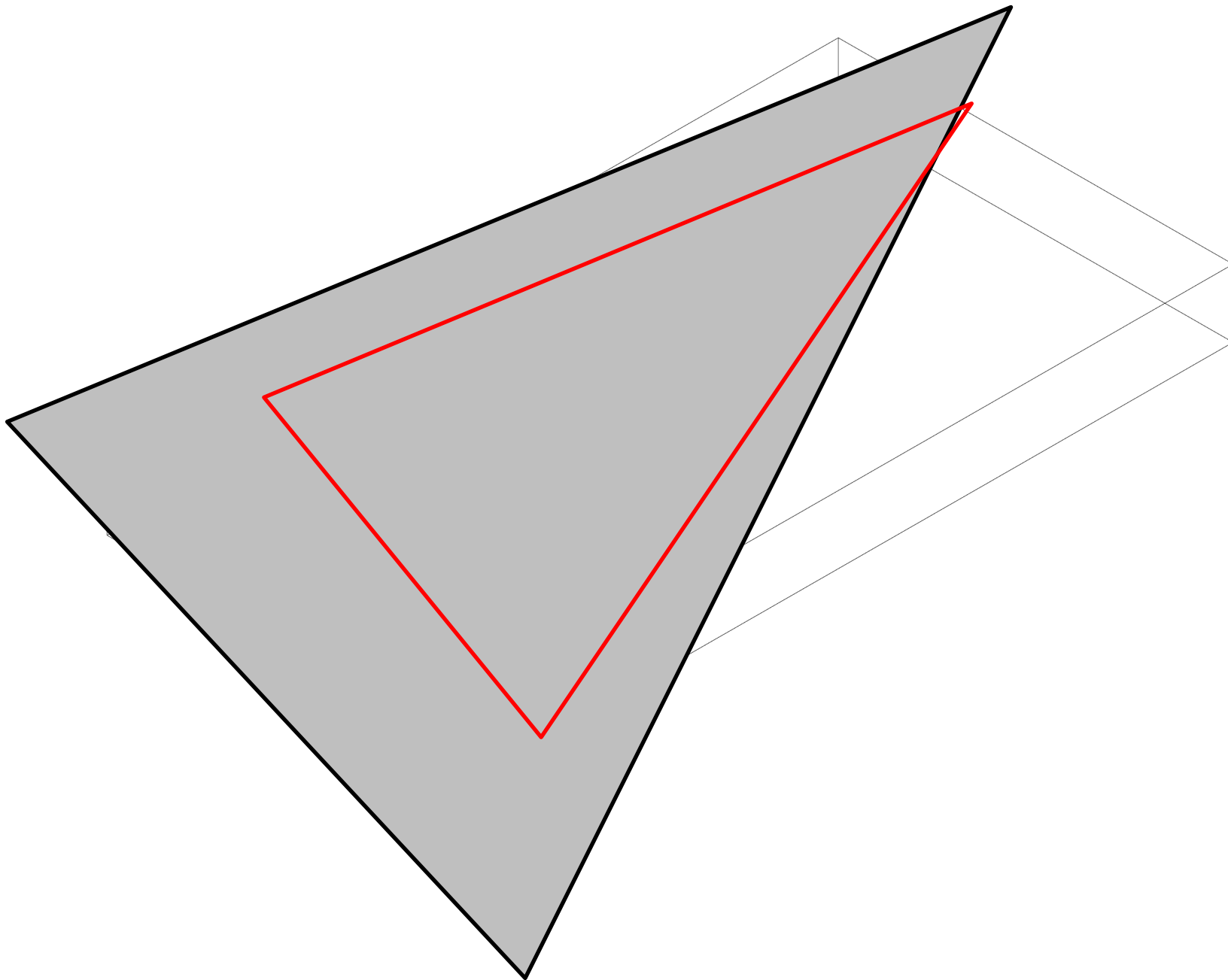




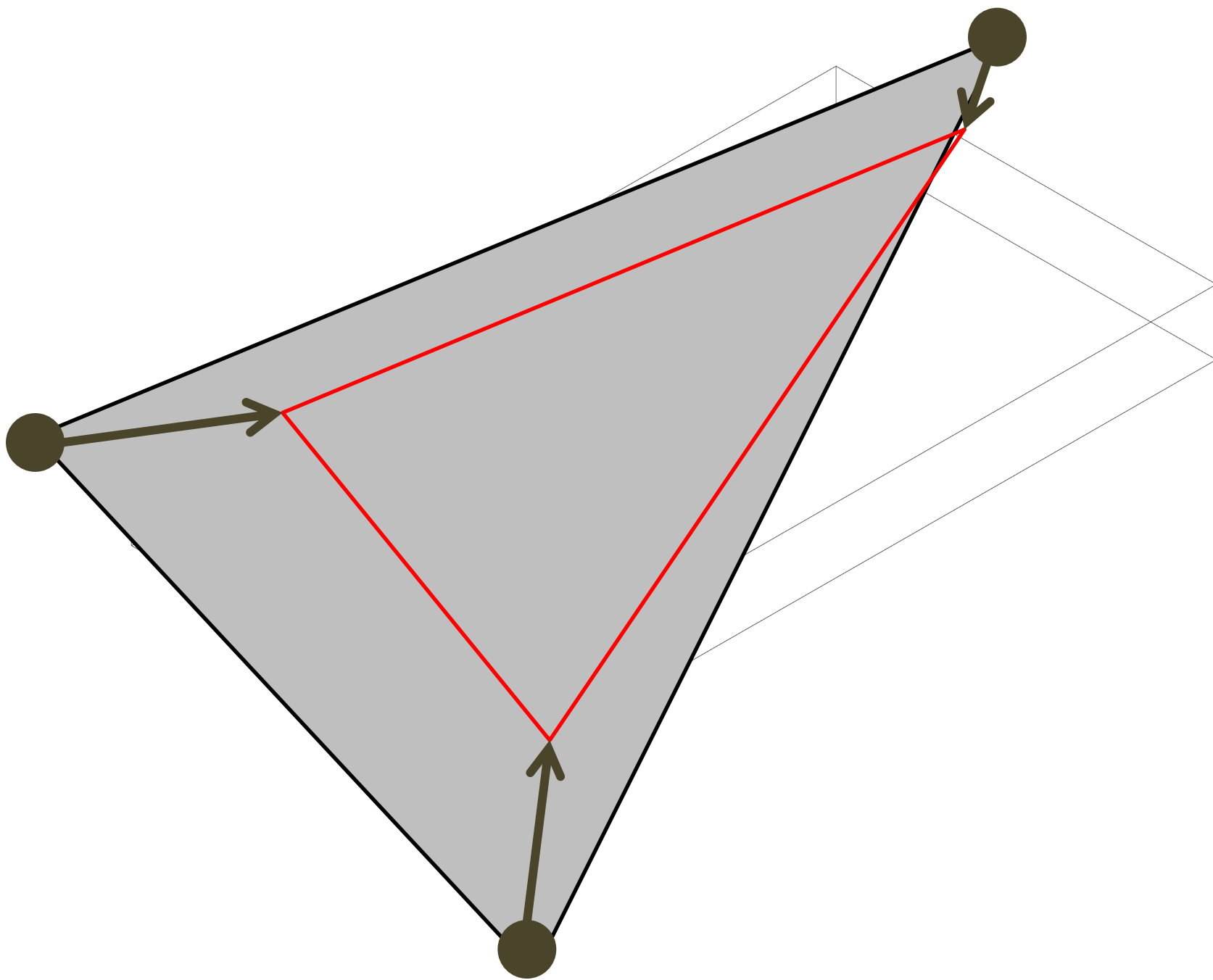






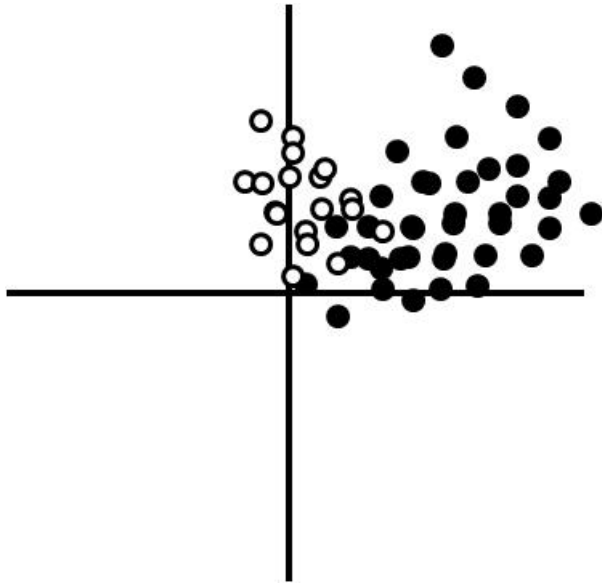




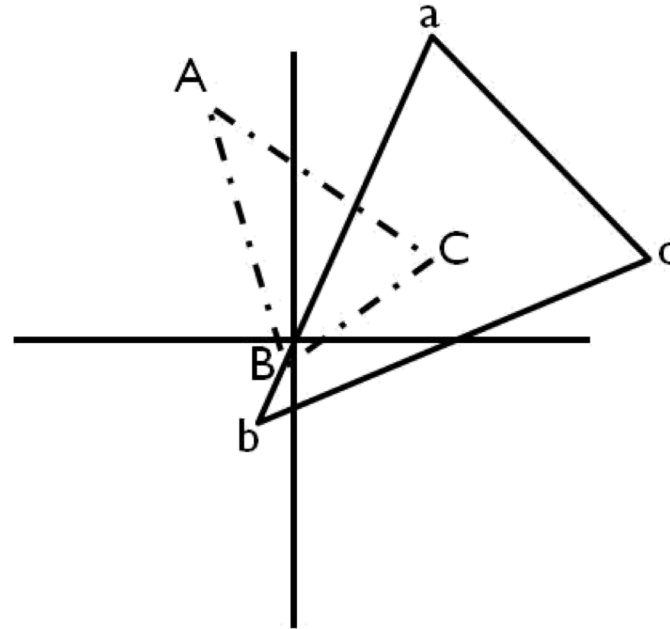


# Análise PROcrustes

Dados originais



Dados originais



**Utiliza um algoritmo que minimiza a soma de quadrados da distância entre dois pontos correspondentes em duas matrizes**

If  $\mathbf{Y}$  is a matrix of Cartesian coordinates:

$$\mathbf{Z} = \left[ \text{tr} \left[ (\mathbf{Y} - \bar{\mathbf{Y}})(\mathbf{Y} - \bar{\mathbf{Y}})^{\text{T}} \right] \right]^{-1/2} (\mathbf{Y} - \bar{\mathbf{Y}}) \mathbf{H}$$

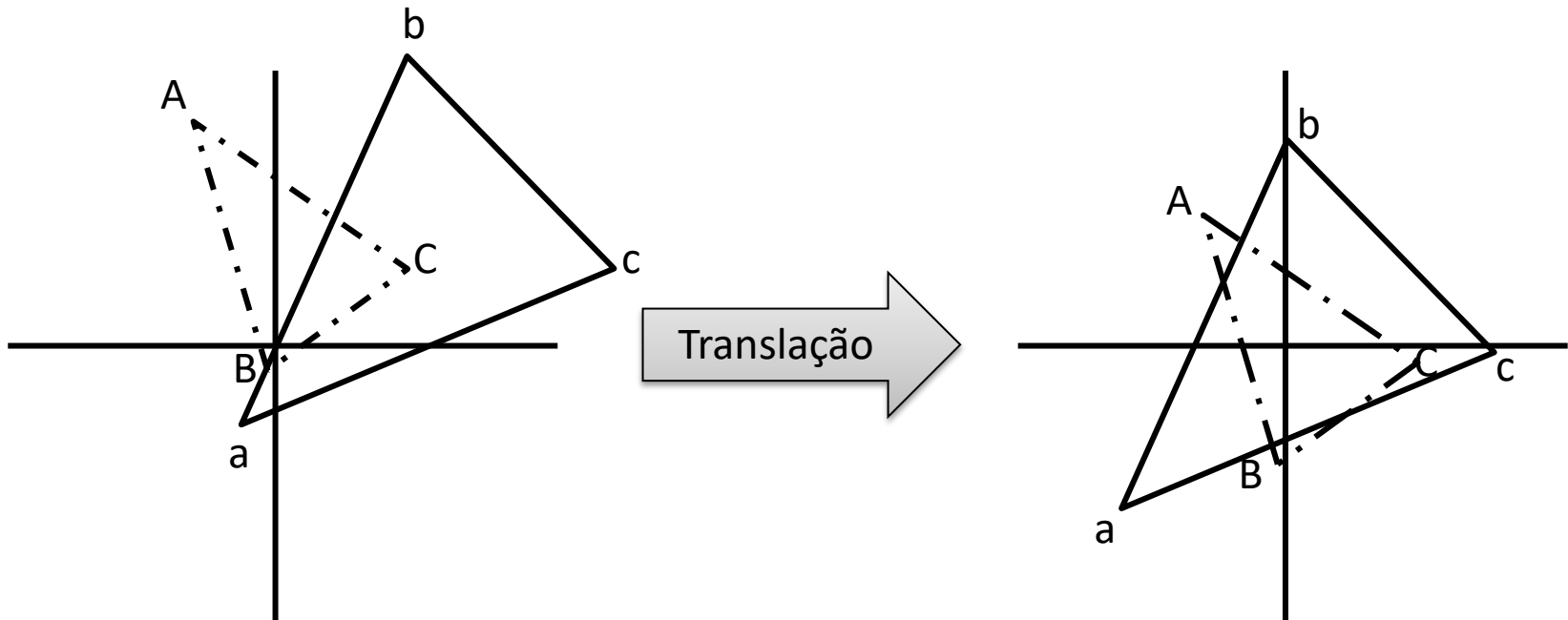
Inverse of Centroid Size  
Rescaling

Centering

Rotation

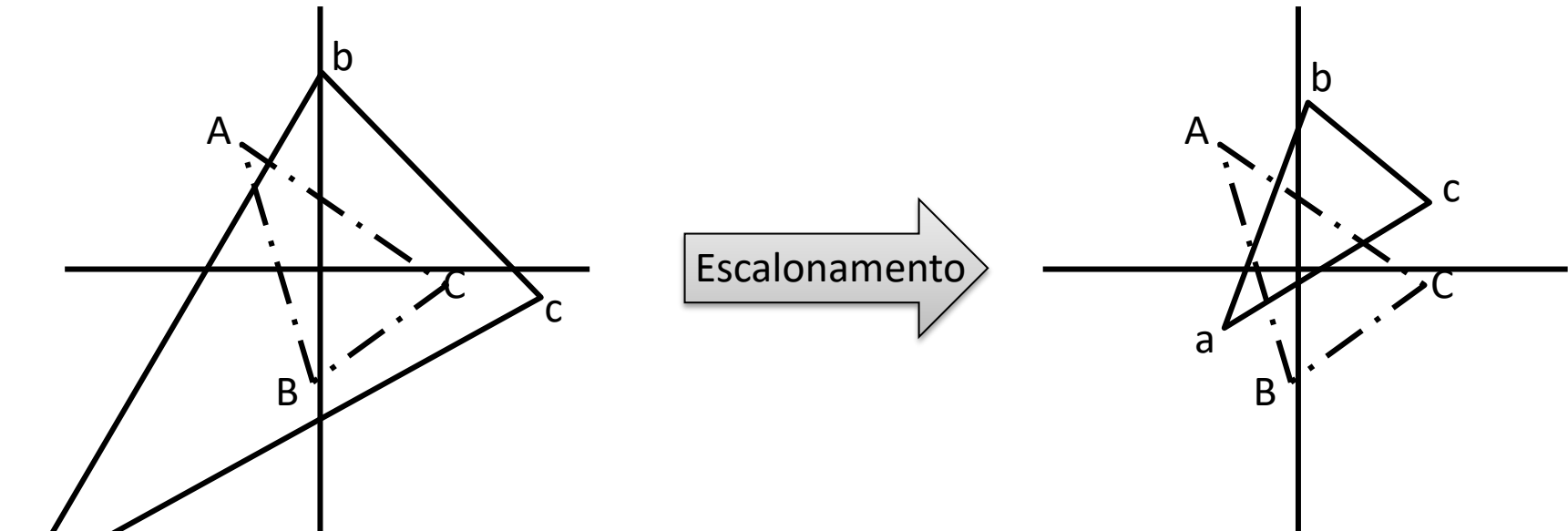
Procrustes  
Residuals

# PROcrustes



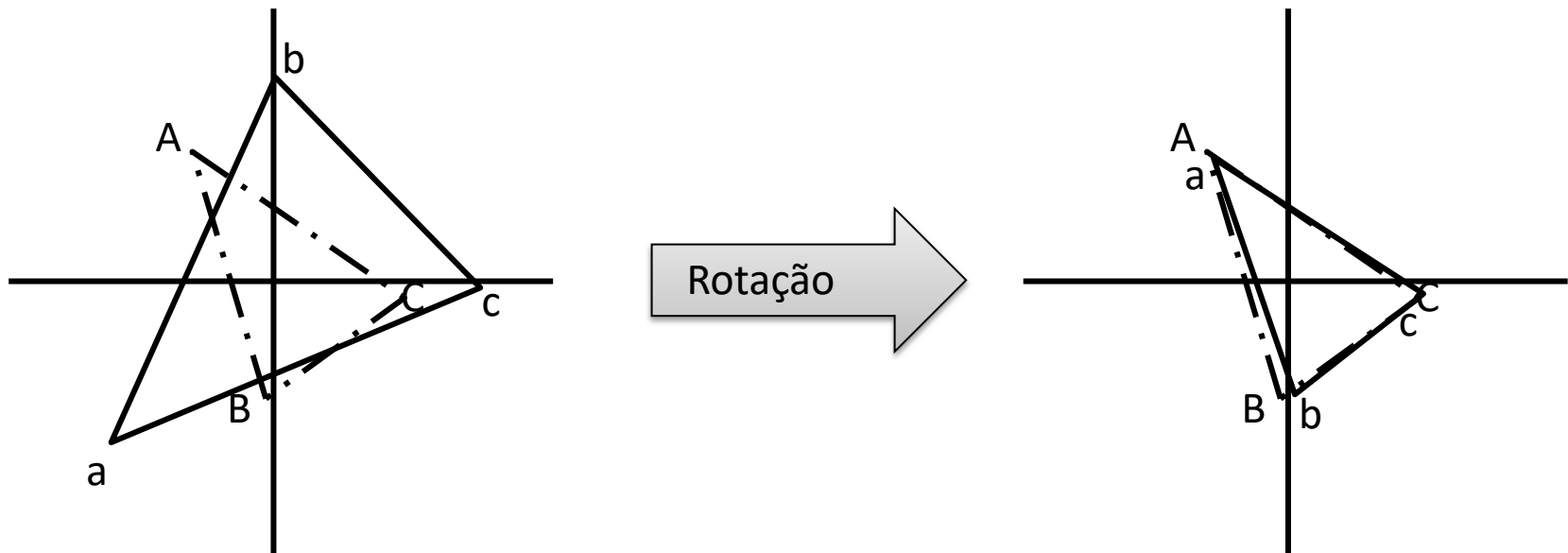
$$\mathbf{Z} = \frac{1}{CS} (\mathbf{Y} - \bar{\mathbf{Y}}) \mathbf{H}$$

# PROcrustes



$$\mathbf{Z} = \frac{1}{CS} (\mathbf{Y} - \bar{\mathbf{Y}}) \mathbf{H}$$

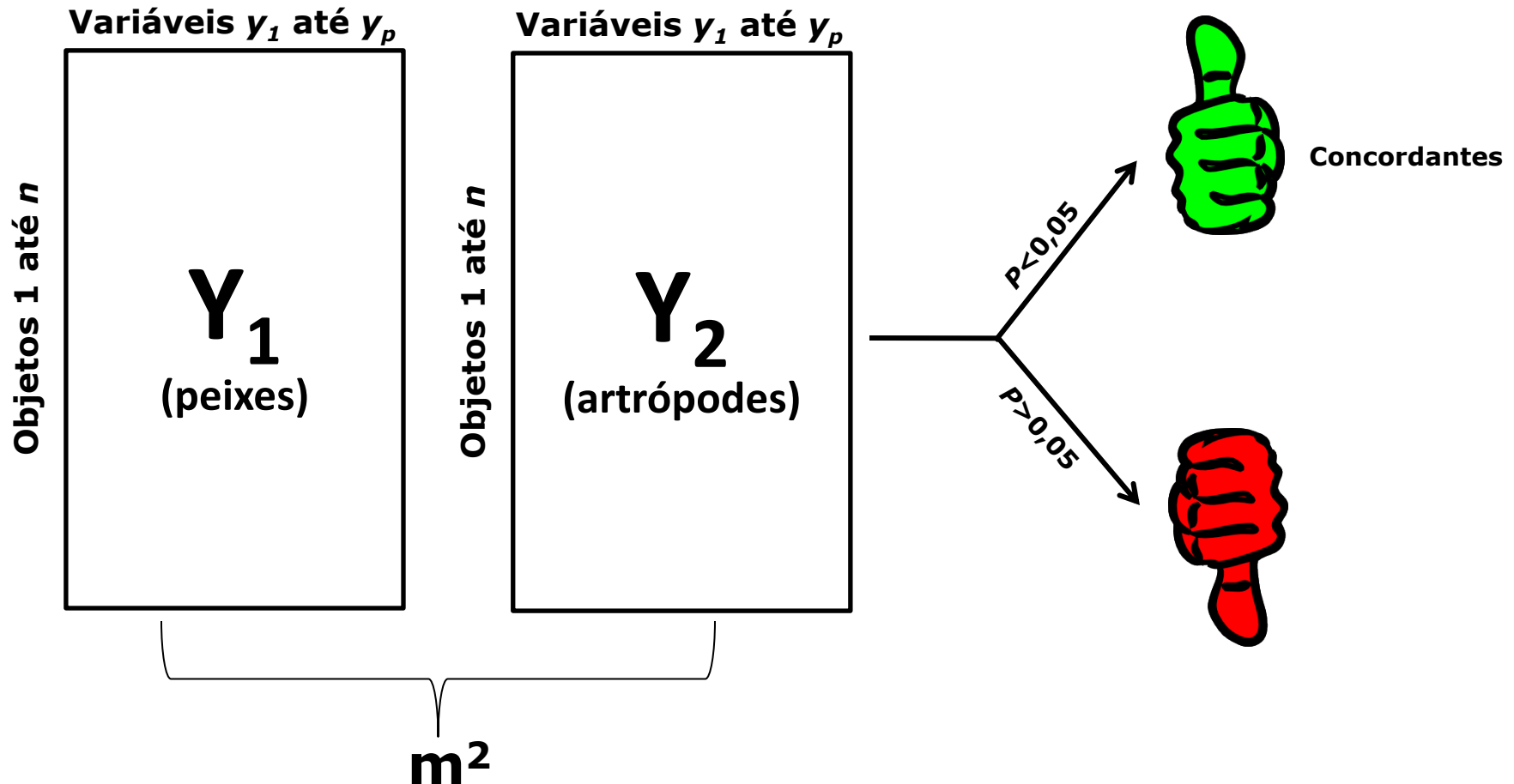
# PROcrustes



$$\mathbf{Z} = \frac{1}{CS} (\mathbf{Y} - \bar{\mathbf{Y}}) \mathbf{H}$$

# Análise PROcrustes

Mede o **grau de concordância** entre duas matrizes



# Estatística do método

$$m_{12}^2 = \text{Trace}(\mathbf{Y}_1 \mathbf{Y}_1') - \frac{(\text{Trace} \mathbf{W})^2}{\text{Trace}(\mathbf{Y}_2 \mathbf{Y}_2')}$$

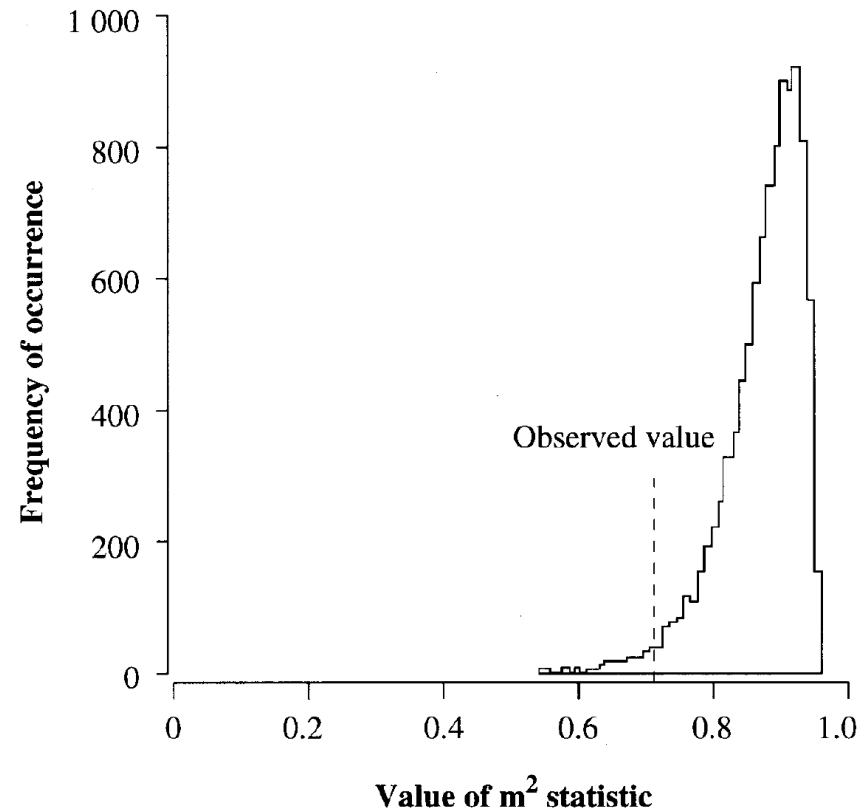


# Teste de hipótese

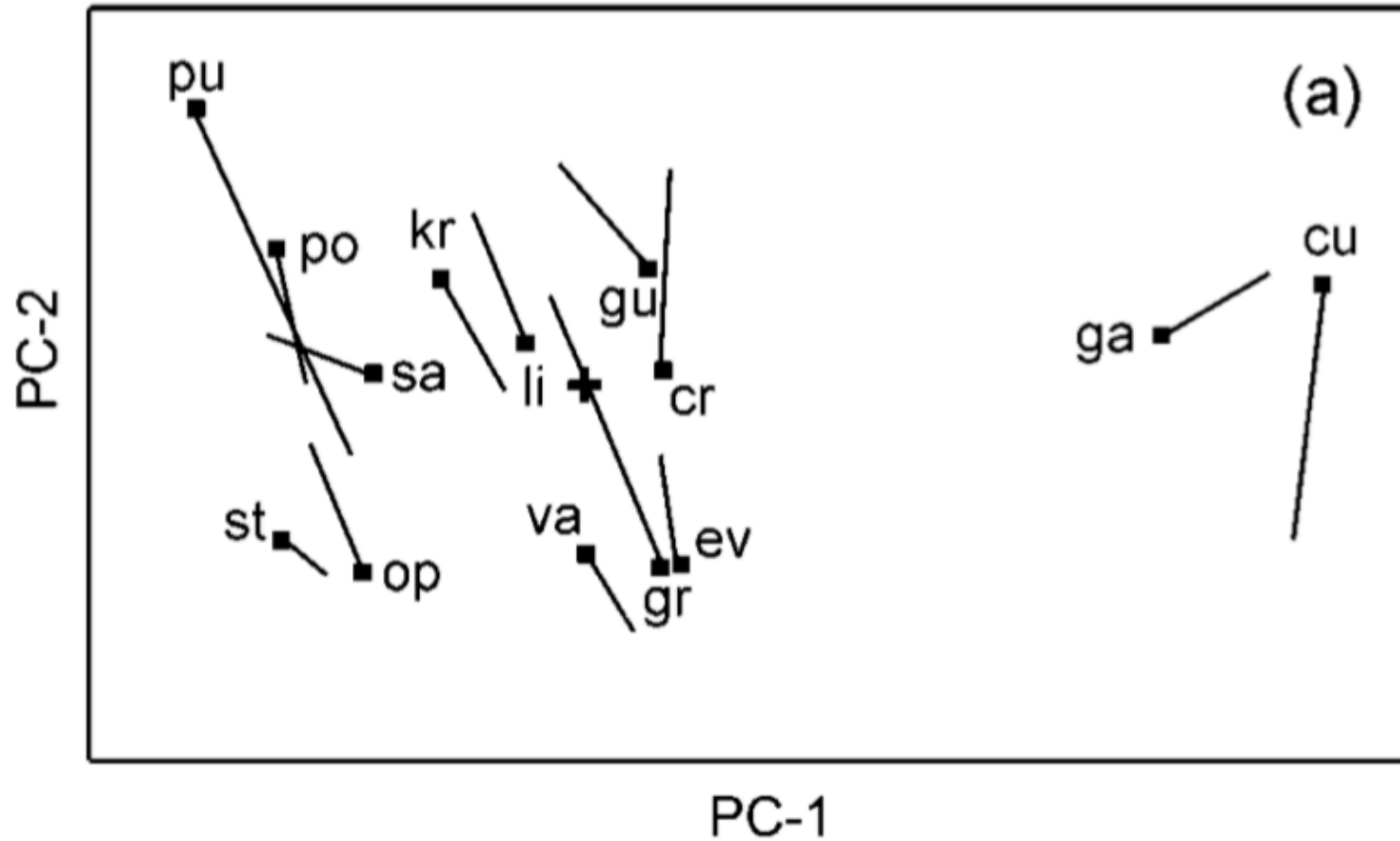
## PROTEST: A PROcrustean Randomization TEST of community environment concordance<sup>1</sup>

Donald A. JACKSON, Department of Zoology, University of Toronto, Toronto, Ontario M5S 1A1, Canada, e-mail: jackson@zoo.utoronto.ca

- PROTEST
  - Teste de hipótese para  $m^2$
  - Baseado em procedimento de aleatorização de Monte Carlo

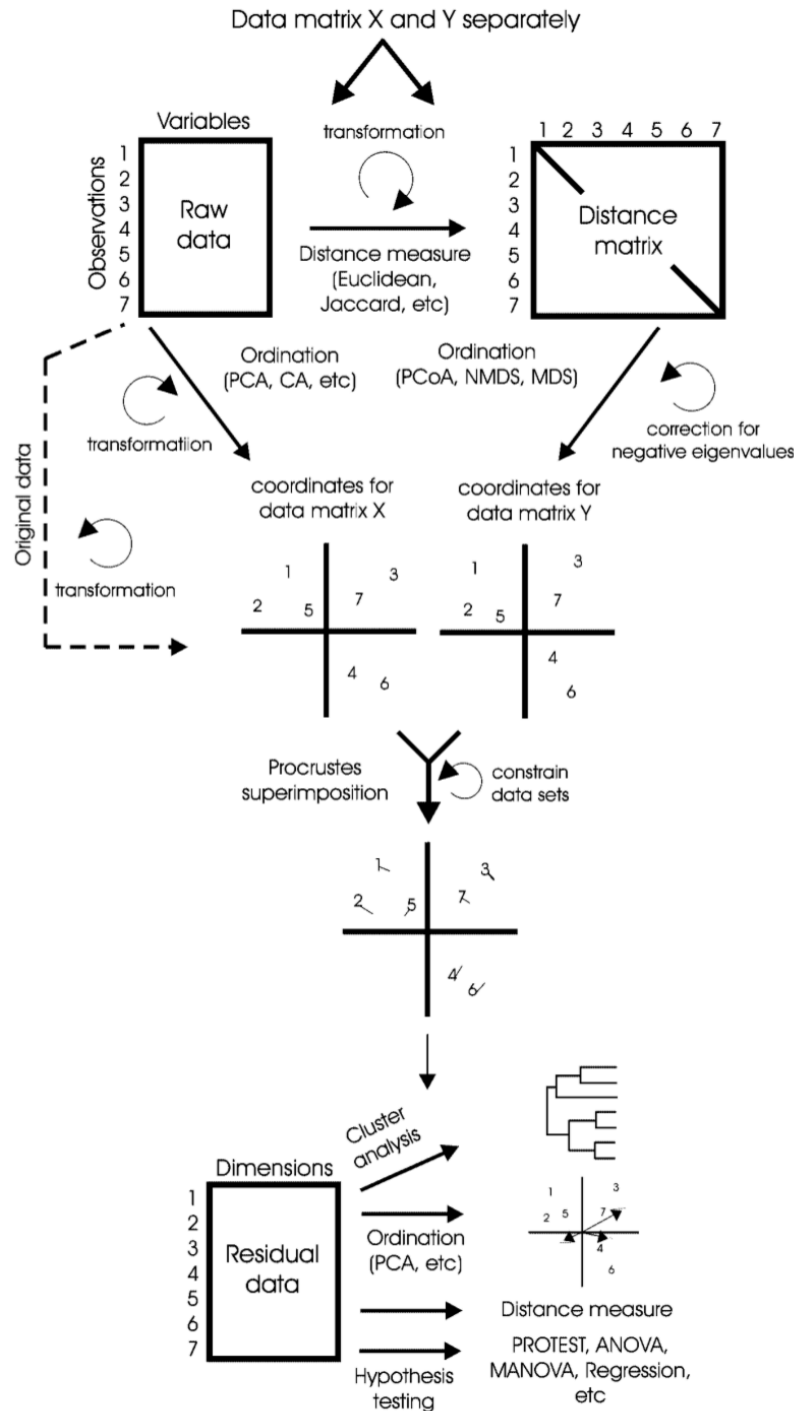


# Exemplo de resultado gráfico



# Vantagens do Procrustes/PROTEST

- Mais poderoso (baixo Erro Tipo I) do que Mantel para detectar padrões
- Não perde informações sobre observações individuais
  - Mantel reduz dados à distância, portanto perde informação sobre os objetos



Testar concordância  
entre conjunto de  
dados

Permite comparar  
ordenações

“Seed” agrupamentos

# PERMANOVA

A test of the general multivariate hypothesis of **differences in the composition** and/or **relative abundances** of organisms of different **species** (variables) in samples from different groups or treatments.

# PERMANOVA

A test of the general multivariate hypothesis of **differences in the composition** and/or **relative abundances** of organisms of different **species** (variables) in samples from different groups or treatments.



Aplicável para múltiplas variáveis resposta (não independentes) medidas em 1 ou mais fatores

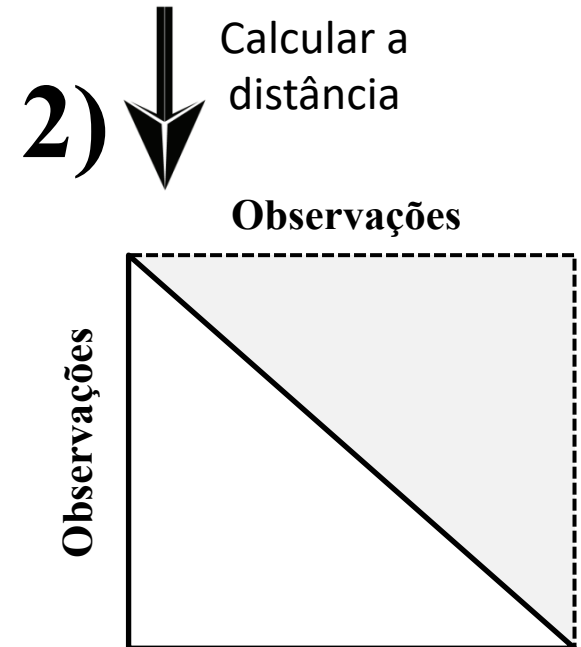
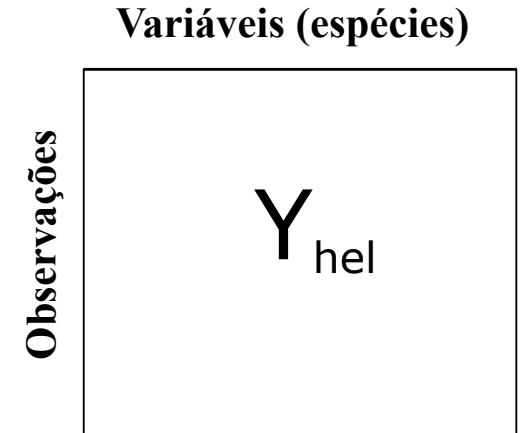
# PERMANOVA

1)



# PERMANOVA

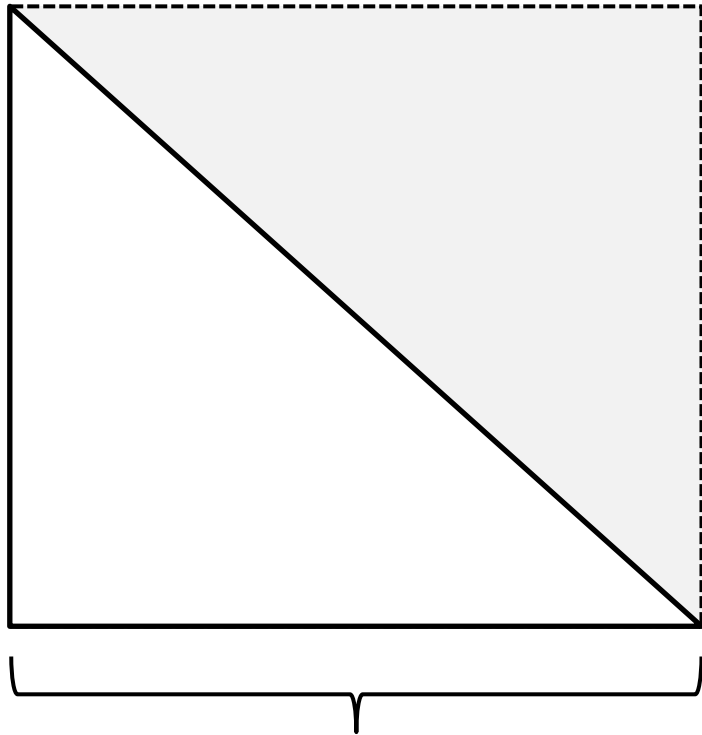
1)





# PERMANOVA (simples – 1 fator)

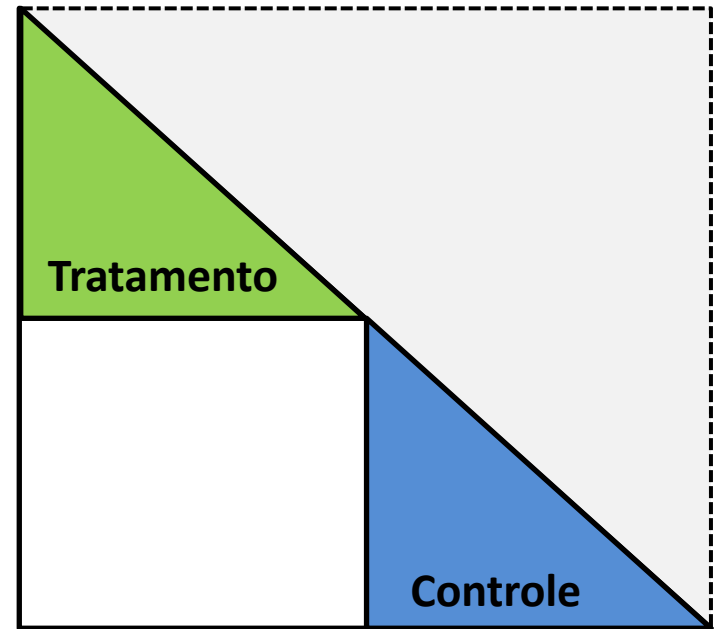
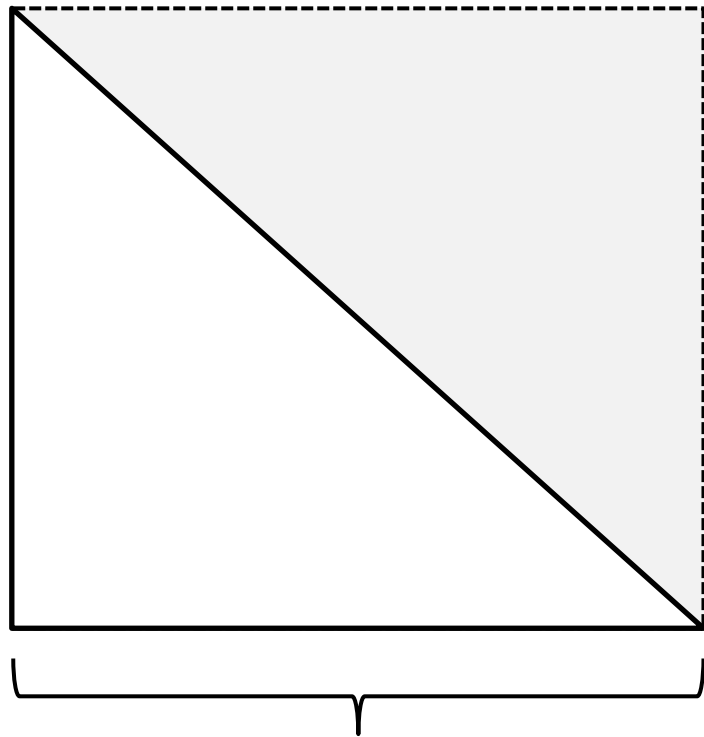
3)



$$SS_T = \left[ \sum_{i=1}^{N-1} \sum_{j=i+1}^N d_{ij}^2 \right] / N$$

# PERMANOVA (simples – 1 fator)

3)

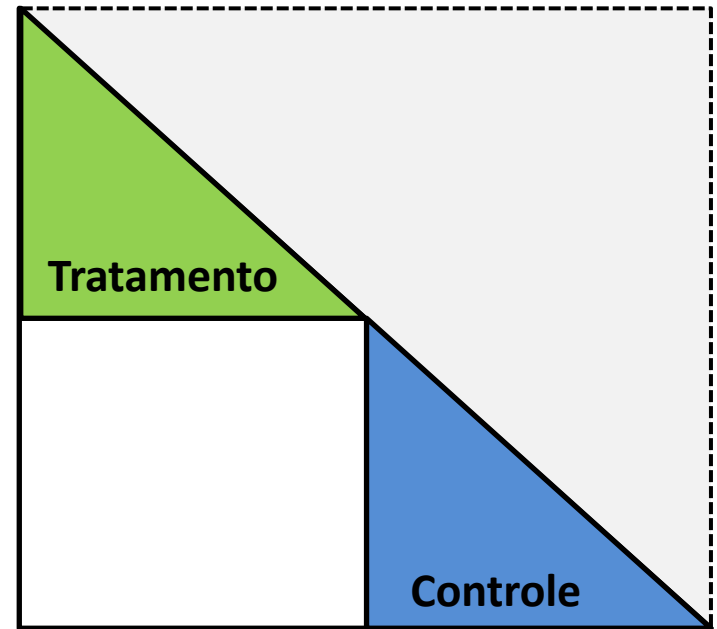
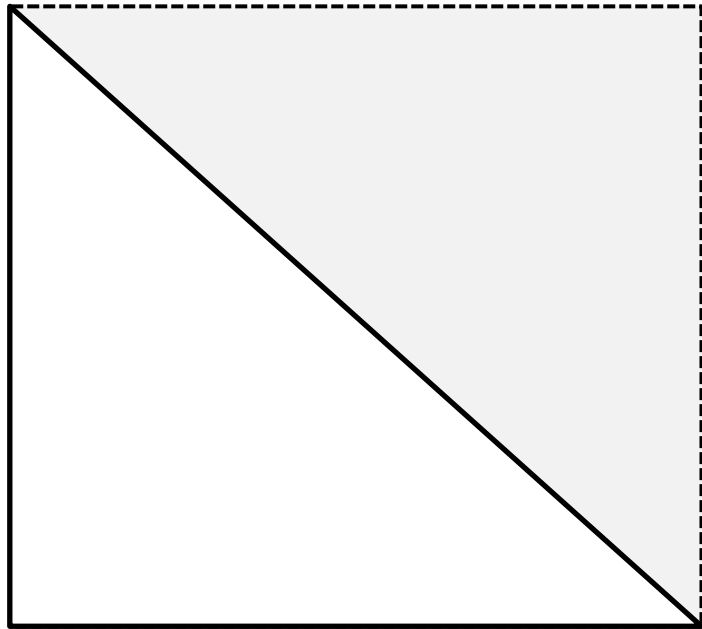


$$SS_T = \left[ \sum_{i=1}^{N-1} \sum_{j=i+1}^N d_{ij}^2 \right] / N$$

$$SS_w = \left[ \sum_{i=1}^{N-1} \sum_{j=i+1}^N \epsilon_{ij} * d_{ij}^2 \right] / n$$

# PERMANOVA (simples – 1 fator)

3)

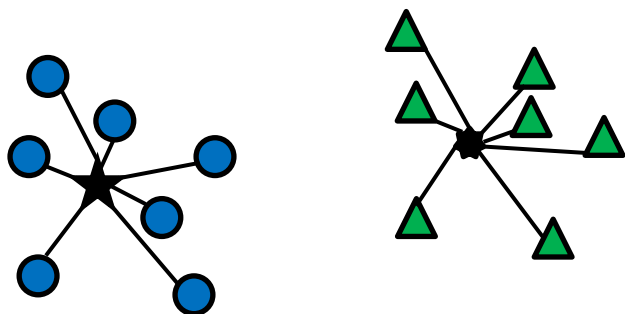


$$SS_A = SS_T - SS_W$$

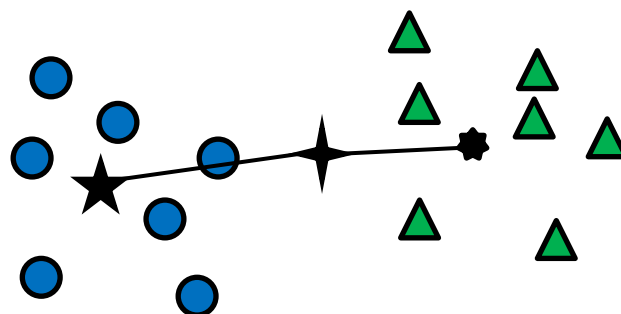
$$F = \frac{SS_A / (a - 1)}{SS_W / (N - a)}$$

a = nº de grupos

**Distância das amostras ao  
centróide do grupo**



**Distância dos centróides do  
grupo ao centróide geral**



Aquecido

Natural

Aquecido

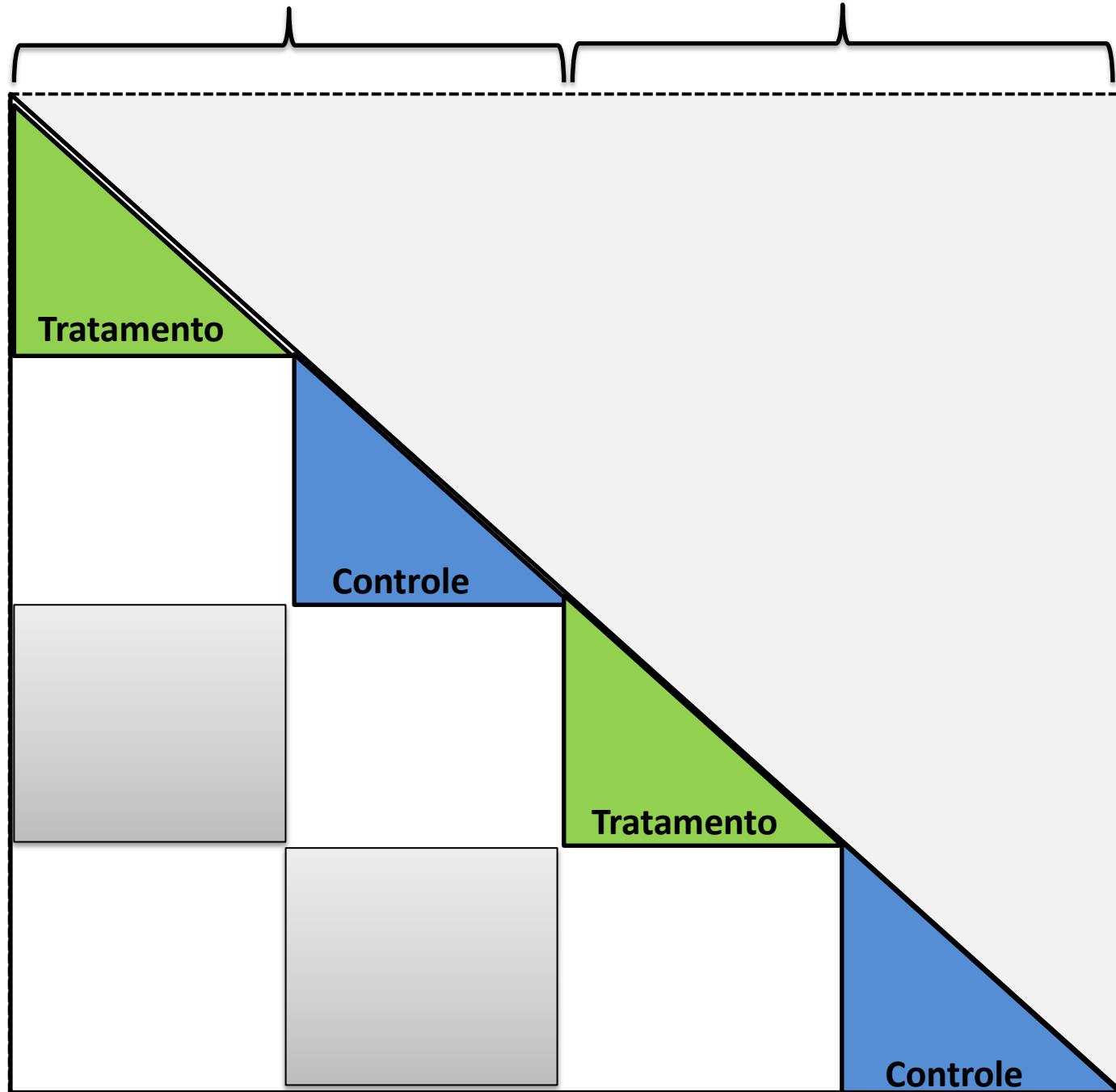
Natural

Tratamento

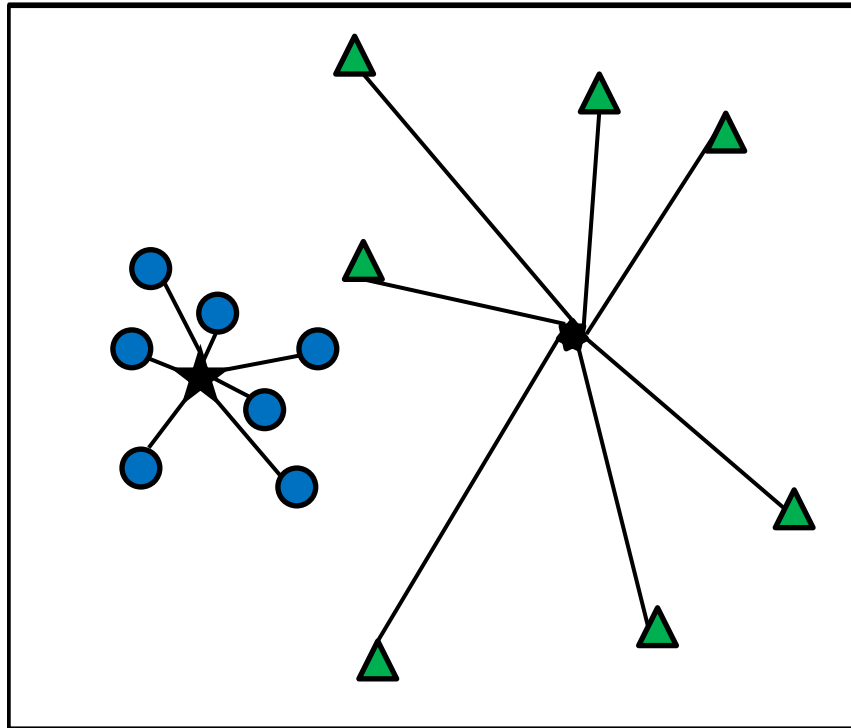
Controle

Tratamento

Controle



**Cuidado! Heterogeneidade nas dispersões**  
- PIOR (ou proibitivo?) PARA ANOSIM E MANTEL



# Tipos de perguntas

Mudança na posição: “Houve uma mudança fundamental na estrutura da comunidade?”

Mudança na dispersão: “a estrutura da comunidade se tornou mais ou menos variável?”

# PERMANOVA, ANOSIM, and the Mantel test in the face of heterogeneous dispersions: What null hypothesis are you testing?

MARTI J. ANDERSON<sup>1,3</sup> AND DANIEL C. I. WALSH<sup>2</sup>



- Para desenhos balanceados, PERMANOVA é mais robusta que ANOSIM e Mantel
- Análises testam diferentes hipóteses nulas
  - ANOSIM e Mantel testam se *amostras no mesmo grupo não são mais agrupadas do que amostras de diferentes grupos*
  - PERMANOVA testa se *existe diferença nos centroides entre grupos*
- Leva em conta a posição do grupo no espaço multivariado e não só a dispersão
- Com MANTEL e ANOSIM não é possível distinguir se a diferença é devido à posição, dispersão, ou formato dos pontos do grupo



- Diogo, preciso confessar, não gostei nada da disciplina, você é um péssimo professor.
- Tudo bem jovem, aqui tem o link pra o curso com o Legendre, com material e vídeos



<http://adn.biol.umontreal.ca/~numerica/ecology/Trieste16/>



*That's all Folks!*

Obrigado pela paciência  
e perseverança!

Dúvidas?



Querem compartilhar o desespero?

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